MACHINE WHITE PAPERS

# Hwacheon Hi-Eco 21 HS — Bearkat Blur

High-speed CNC lathe for small-part precision turning where accuracy and cycle time matter most. 8" OD × 12" L max turning.




**Bearkat Blur** · high-speed precision CNC turning center

Bearkat Blur exists because certain oilfield and aerospace parts are small, high-quantity, and can't tolerate the setup time of a larger machine. Bushings, couplings, precision shafts, valve inserts, and any repeatable geometry in the 8-inch envelope — this machine cycles fast without giving up accuracy. When cycle time drives the quote, this is the machine that wins.

## What Bearkat Blur is for

Bearkat Blur is the shop's answer to a specific economic problem: some oilfield and aerospace parts are small, wanted in quantity, and cannot carry the setup time of a larger lathe. 8" OD × 12" L, a 2" through-spindle bar capacity, 4,000 RPM, and a 12-station turret with 1,181 IPM rapids on both axes — every one of those numbers is about cycle time. When the quote is won or lost on price per piece at quantity 200, this is the machine that wins it.

### Pick this machine when

- The part is under 8" in diameter and 12" long — bushings, couplings, valve inserts, precision shafts, seat rings.
- Quantity is high enough that seconds per piece show up in the price.
- Bar-fed work up to 2" through the spindle.
- The part is small but the tolerance isn't loose — this machine holds ±0.0005" while cycling fast.

> **When it's the wrong machine**
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> Anything that doesn't fit. The envelope is a hard wall, and the honest answer on a 9-inch part is that it goes to a Hi-Eco 35 cell instead. Heavy interrupted roughing on forged stock is also wrong here — 15 HP continuous is sized for finish-critical small work, not for taking scale off a forging. That's the Hammer of Forge's job.

### From the floor

Small parts in Inconel 718 and 17-4 PH punish tool wear more than large ones, because the same insert edge is doing proportionally more of the part. On repeat runs the insert-change interval gets set from measured wear on the first batch rather than from a catalogue number, and finish-critical features get a fresh edge regardless of where that interval lands.

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## Pick your industry

### Hwacheon Hi-Eco 21 HS for Oil & Gas

Full spec sheet, typical oil & gas parts, alloys, tolerance expectations, documentation approach.

### Hwacheon Hi-Eco 21 HS for Aerospace

Full spec sheet, typical aerospace parts, alloys, tolerance expectations, documentation approach.

### Hwacheon Hi-Eco 21 HS for Military & Defense

Full spec sheet, typical defense parts, alloys,

### Hwacheon Hi-Eco 21 HS for Industrial & General Manufacturing

Full spec sheet, typical industrial parts, alloys,

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## Frequently asked

**What does B&R Productions use the Hwacheon Hi-Eco 21 HS for?**

**When is the Bearkat Blur the right machine for my part?**

**When is the Bearkat Blur the wrong machine?**

**What industries does B&R machine on the Hwacheon Hi-Eco 21 HS?**

**How do I get a quote for work on the Hwacheon Hi-Eco 21 HS?**

Have a project?